EXPERIMENT:9 EXPERIMENT TO UNDERSTAND MATPLOTLIB LIBRARY USE CASES IN DATA SCIENCE THROUGH VISUALIZATION

CODE :

```
import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

%matplotlib inline

data = pd.read_csv("Iris_Dataset.csv")

print(data.info())

print(data.describe())

print(data.value_counts('variety'))

sns.countplot(x='variety', data=data)

plt.show()

dummies = pd.get_dummies(data.variety)

FinalDataset = pd.concat([dummies, data.iloc[:, [0,1,2,3]]], axis=1)

print(FinalDataset.head())

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data)

plt.show()

sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data)

plt.show()
```

```
sns.pairplot(data, hue='variety', height=3)

plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'PetalLengthCm').add_legend()

plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'SepalLengthCm').add_legend()

plt.show()

sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot,
'SepalWidthCm').add_legend()

plt.show()
```

OUTPUT :

```
<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

 #   Column        Non-Null Count  Dtype

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 0   Id            150 non-null    int64

 1   SepalLengthCm  150 non-null    float64

 2   SepalWidthCm   150 non-null    float64

 3   PetalLengthCm  150 non-null    float64

 4   PetalWidthCm   150 non-null    float64
```

5 variety 150 non-null object

dtypes: float64(4), int64(1), object(1)

memory usage: 7.2+ KB

None

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.057333	3.758000	1.199333
std	43.445368	0.828066	0.435866	1.765298	0.762238
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

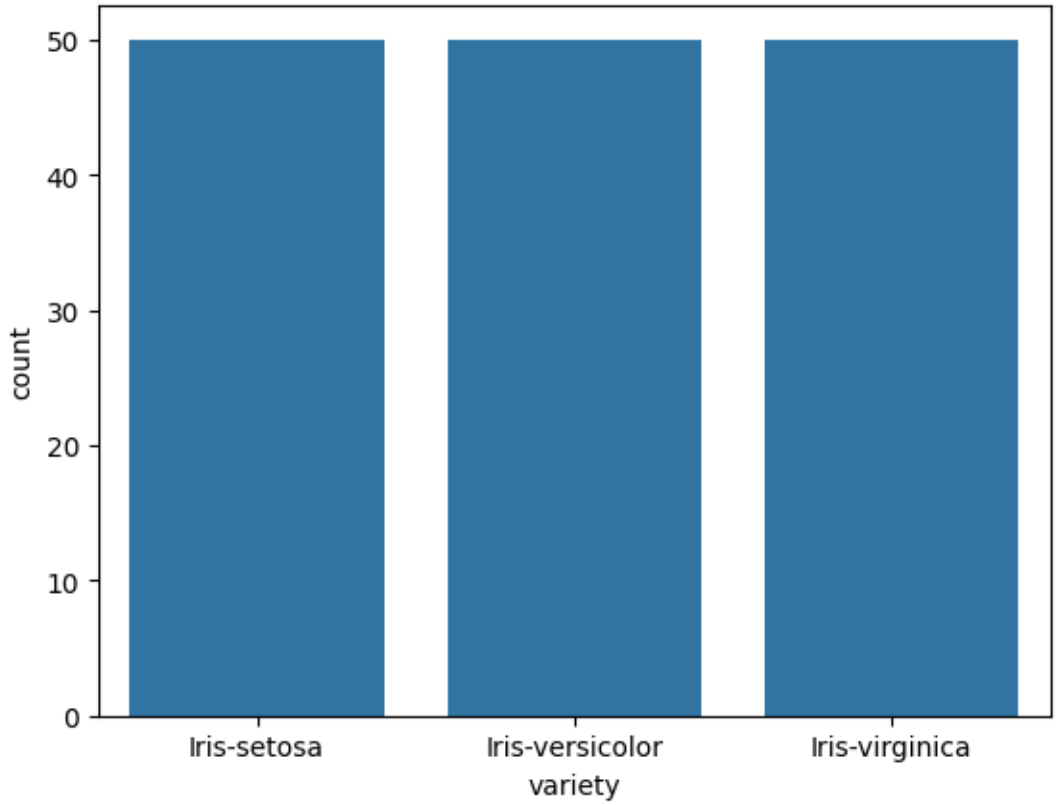
variety

Iris-setosa 50

Iris-versicolor 50

Iris-virginica 50

Name: count, dtype: int64

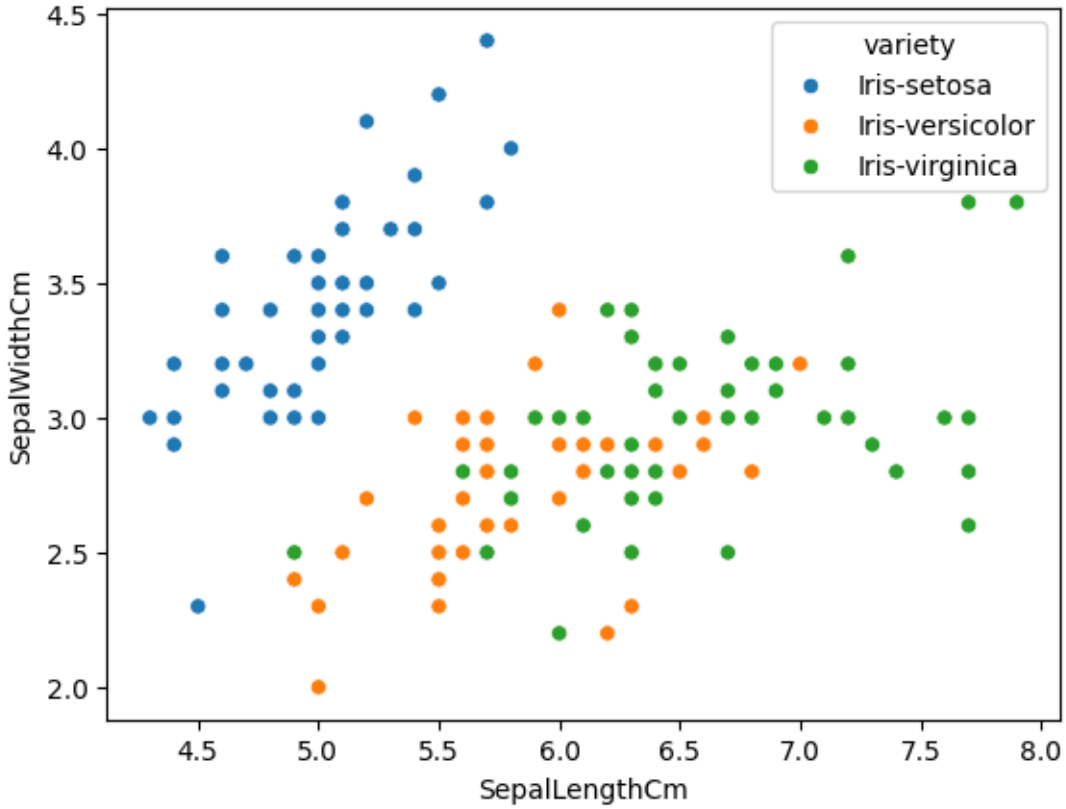


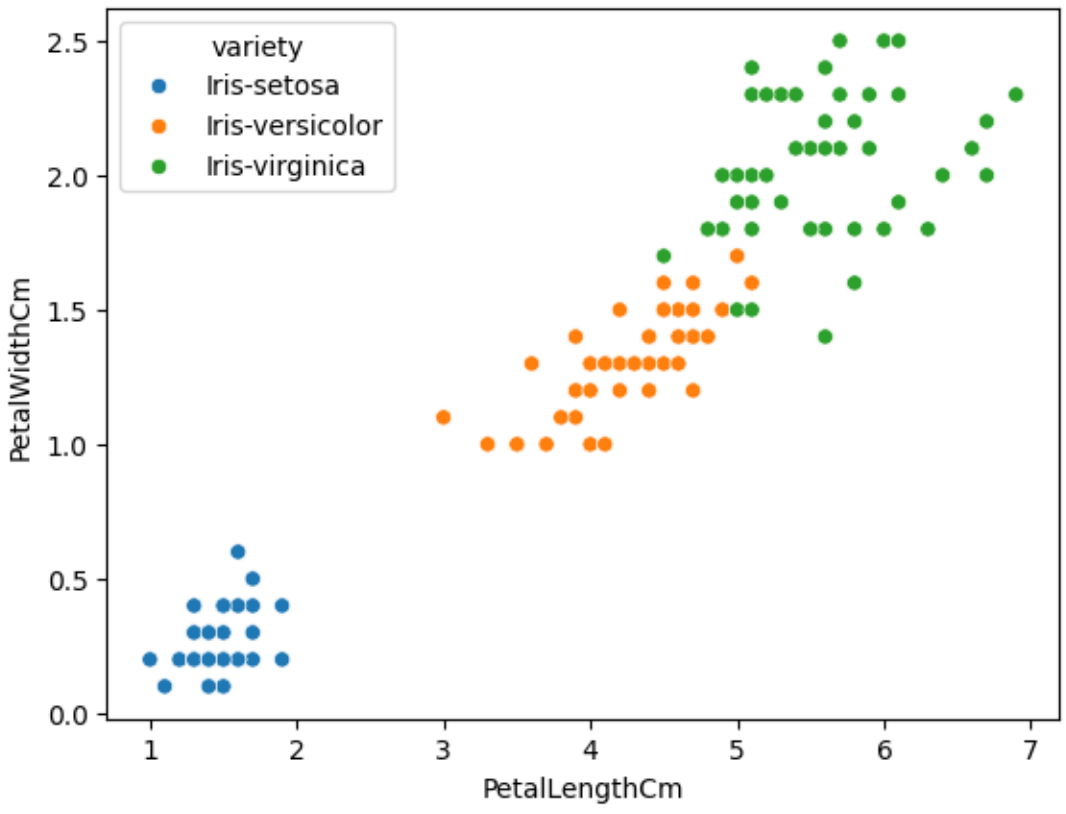
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	\
0	True	False	False	1	5.1	
1	True	False	False	2	4.9	
2	True	False	False	3	4.7	
3	True	False	False	4	4.6	
4	True	False	False	5	5.0	

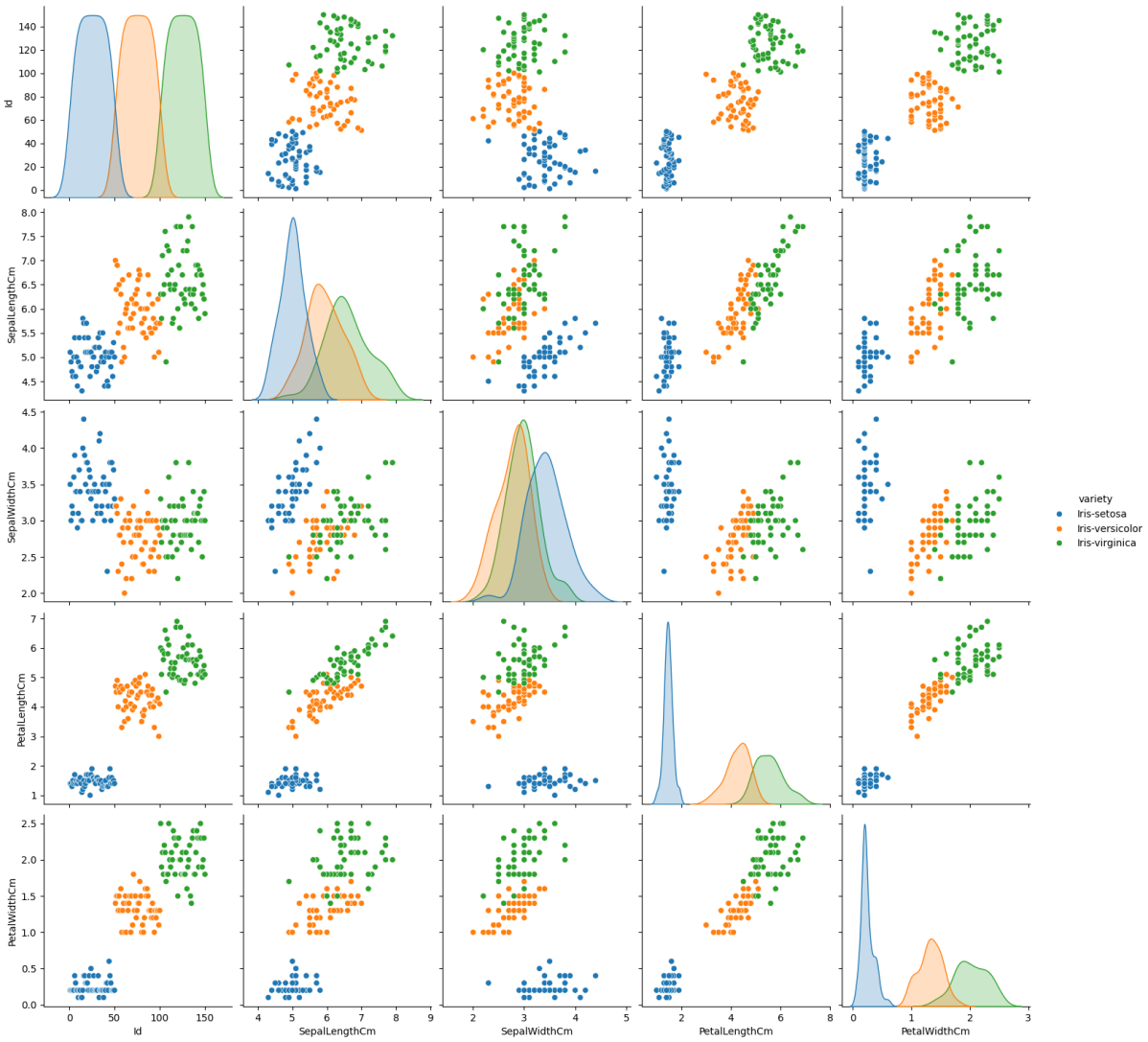
	SepalWidthCm	PetalLengthCm
0	3.5	1.4
1	3.0	1.4
2	3.2	1.3

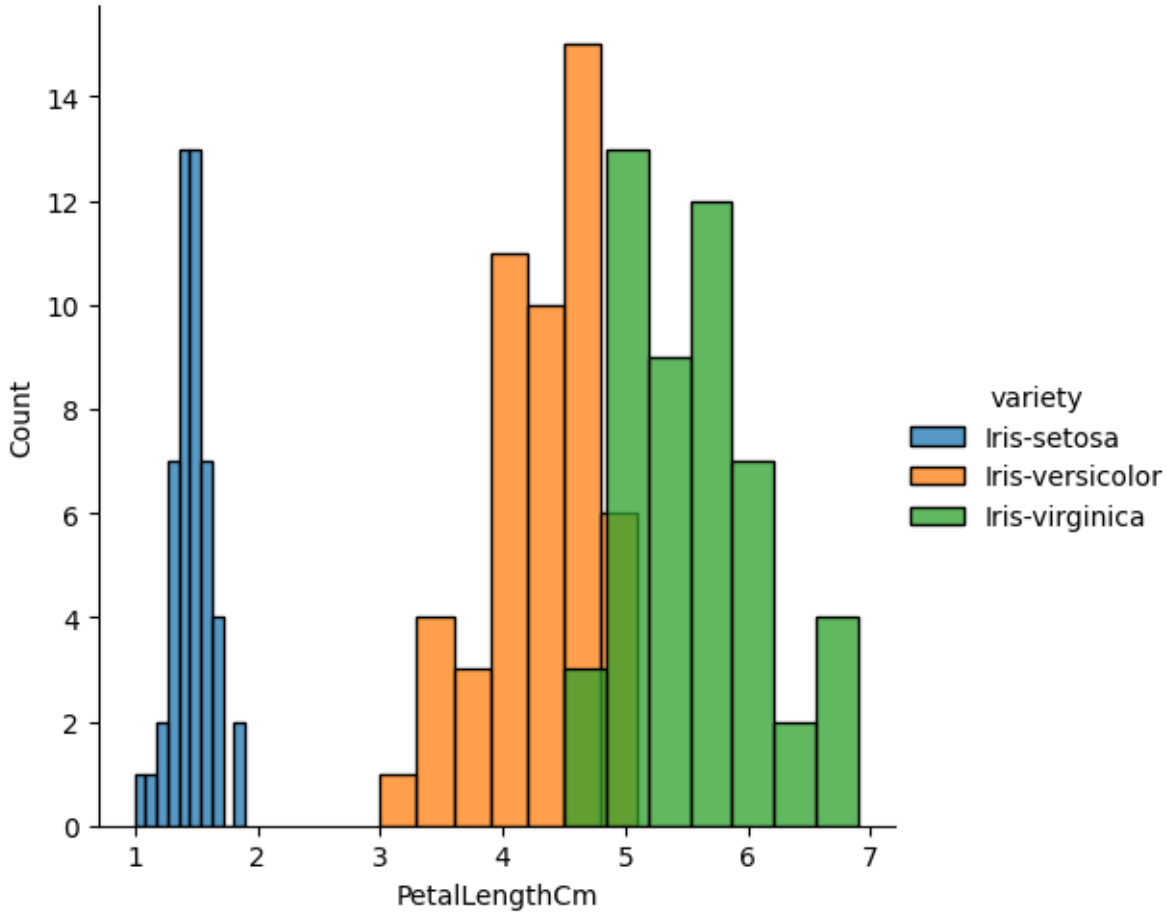
3 3.1 1.5

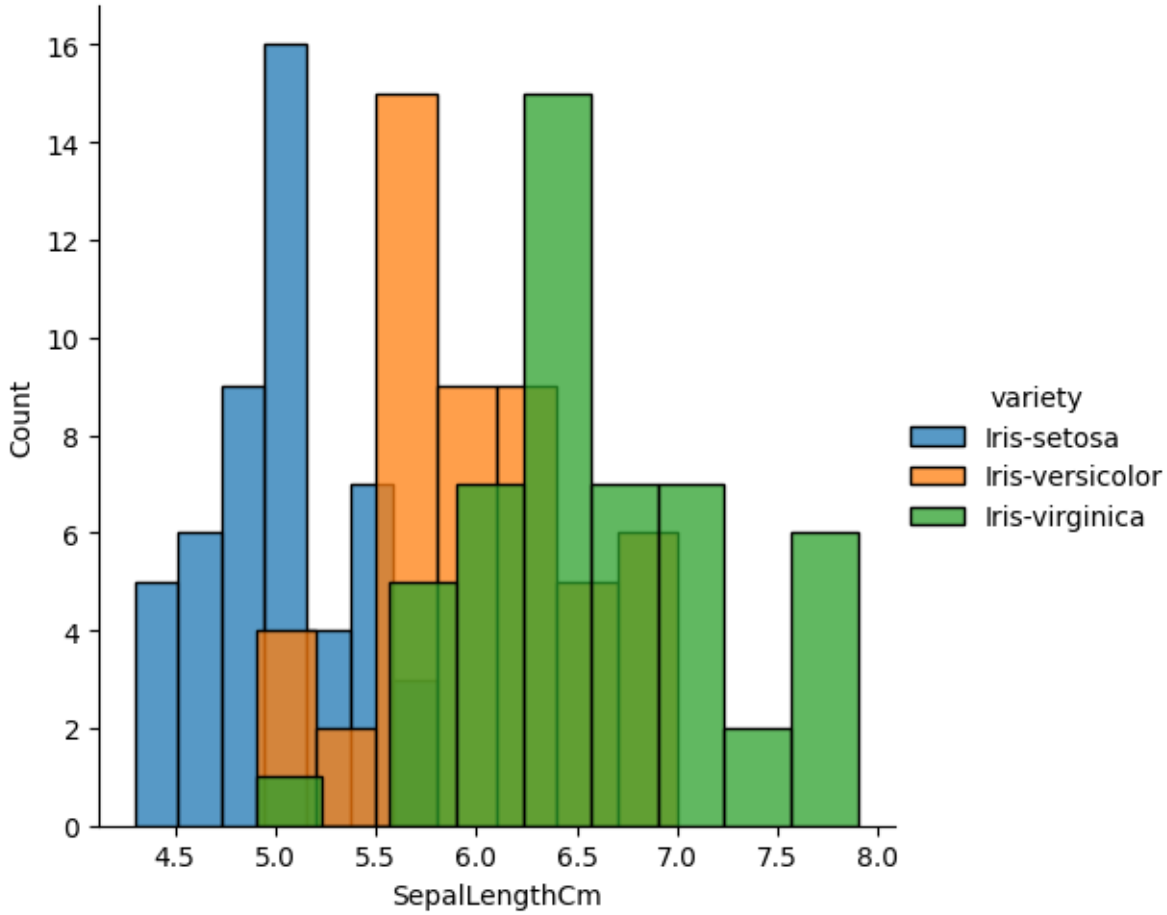
4 3.6 1.4

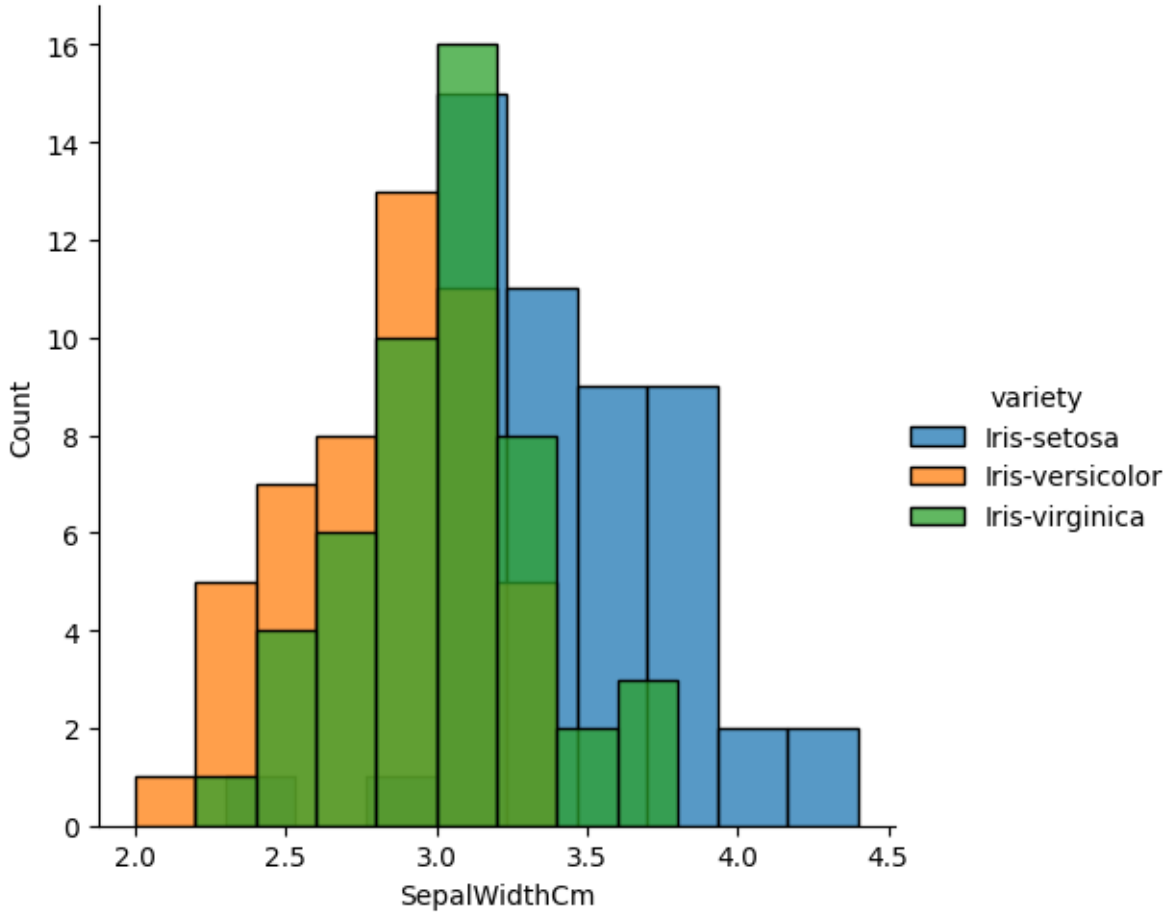

variety
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7


Id
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
SepalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica